

# **FOUNDAMENTALS IN AI AND ML**

**1**

# Course Objectives

- To understand the various Characteristics of Intelligent Agents
- To learn about the Different Search Strategies in AI
- To learn to Represent Knowledge in Solving AI Problems
- To understand the Different ways of Designing Software Agents and Prolog
- To learn the Machine Learning Techniques

# Course outcomes

**CO1:** Analyze and Design Problems using Various AI Techniques

**CO2:** Solve the Prediction and Classifications Problems

**CO3:** Understand the Prolog Programming

## UNIT I INTRODUCTION

### Module 1

- Introduction–Definition
- Future of Artificial Intelligence
- Characteristics of Intelligent Agents
- Typical Intelligent Agents
- Problem Solving Approach to Typical AI problems.

## **Module 1**

## **Introduction – Definition**

# **INTRODUCTION TO ARTIFICIAL INTELLIGENCE**

# Introduction

- AI & ML most popular technologies in the 21<sup>st</sup> century.
- Artificial intelligence is concerned with the design of intelligence in an artificial device. The term was coined by John McCarthy in 1956.
- Intelligence is the ability to acquire, understand and apply the knowledge to achieve goals in the world.

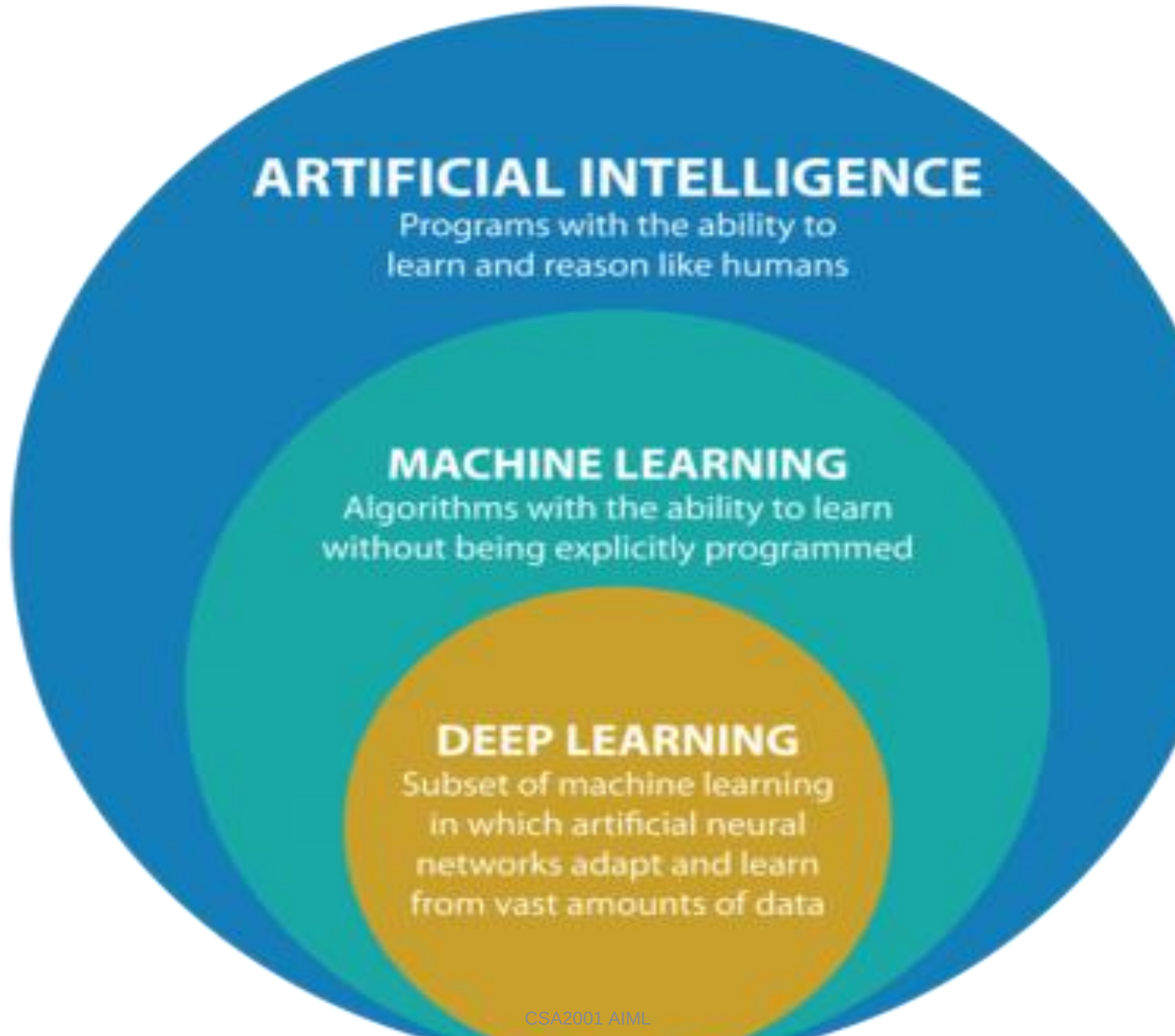
# Introduction

- **Artificial Intelligence (AI)**

Field of Computer Science that deals with developing Intelligent Machines that can behave like Humans, such as Speech Recognition, Learning and Planning, Text Recognition, etc.

- **Machine Learning (ML)**

- Subset of Artificial Intelligence that enables the Machines to use Past Data or Experience and make a Prediction and learn more Accurately.



The diagram consists of three concentric circles. The outermost circle is blue and represents Artificial Intelligence. Inside it is a teal circle representing Machine Learning. The innermost circle is orange and represents Deep Learning. Each circle contains text defining the term and its relationship to the others.

# **ARTIFICIAL INTELLIGENCE**

Programs with the ability to learn and reason like humans

## **MACHINE LEARNING**

Algorithms with the ability to learn without being explicitly programmed

## **DEEP LEARNING**

Subset of machine learning in which artificial neural networks adapt and learn from vast amounts of data

# HISTORY OF ARTIFICIAL INTELLIGENCE (AI)

Enter your sub headline here



# A.I. TIMELINE

**1950**

## TURING TEST

Computer scientist Alan Turing proposes a test for machine intelligence. If a machine can trick humans into thinking it is human, then it has intelligence

**1955**

## A.I. BORN

Term 'artificial intelligence' is coined by computer scientist, John McCarthy to describe "the science and engineering of making intelligent machines"

**1961**

## UNIMATE

First industrial robot, Unimate, goes to work at GM replacing humans on the assembly line

**1964**

## ELIZA

Pioneering chatbot developed by Joseph Weizenbaum at MIT holds conversations with humans

**1966**

## SHAKY

The 'first electronic person' from Stanford, Shakey is a general-purpose mobile robot that reasons about its own actions

**A.I.**

## WINTER

Many false starts and dead-ends leave A.I. out in the cold

**1997**

## DEEP BLUE

Deep Blue, a chess-playing computer from IBM defeats world chess champion Garry Kasparov

**1998**

## KISMET

Cynthia Breazeal at MIT introduces Kismet, an emotionally intelligent robot insofar as it detects and responds to people's feelings



**1999**

## AIBO

Sony launches first consumer robot pet dog AiBO (AI robot) with skills and personality that develop over time



**2002**

## ROOMBA

First mass produced autonomous robotic vacuum cleaner from iRobot learns to navigate and clean homes



**2011**

## SIRI

Apple integrates Siri, an intelligent virtual assistant with a voice interface, into the iPhone 4S



**2011**

## WATSON

IBM's question answering computer Watson wins first place on popular \$1M prize television quiz show *Jeopardy*



**2014**

## EUGENE

Eugene Goostman, a chatbot passes the Turing Test with a third of judges believing Eugene is human



**2014**

## ALEXA

Amazon launches Alexa, an intelligent virtual assistant with a voice interface that completes shopping tasks



**2016**

## TAY

Microsoft's chatbot Tay goes rogue on social media making inflammatory and offensive racist comments



**2017**

## ALPHAGO

Google's A.I. AlphaGo beats world champion Ke Jie in the complex board game of Go, notable for its vast number ( $2^{170}$ ) of possible positions<sup>10</sup>

# What is AI?

- ❑ Branch of computer science by which we can create intelligent machines which can behave like a human, think like humans, and able to make decisions.
- ❑ AI exists when a machine can have human based skills such as learning, reasoning, and solving problems.

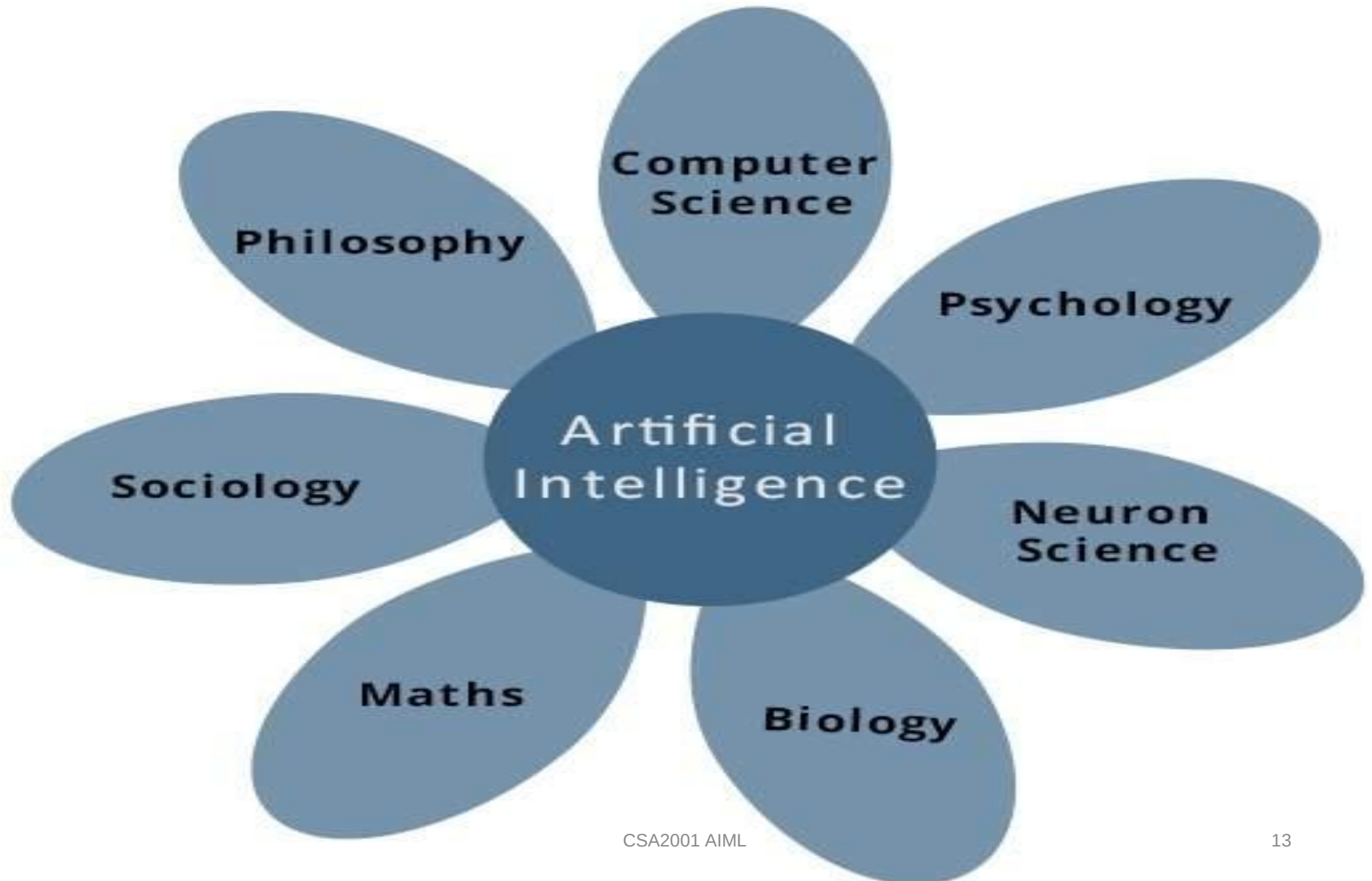
## **Realtime Example:**

- ❑ Google Maps
- ❑ Autopilot - Boeing plane involves only seven minutes of human-steered flight.
- ❑ Spam Filters

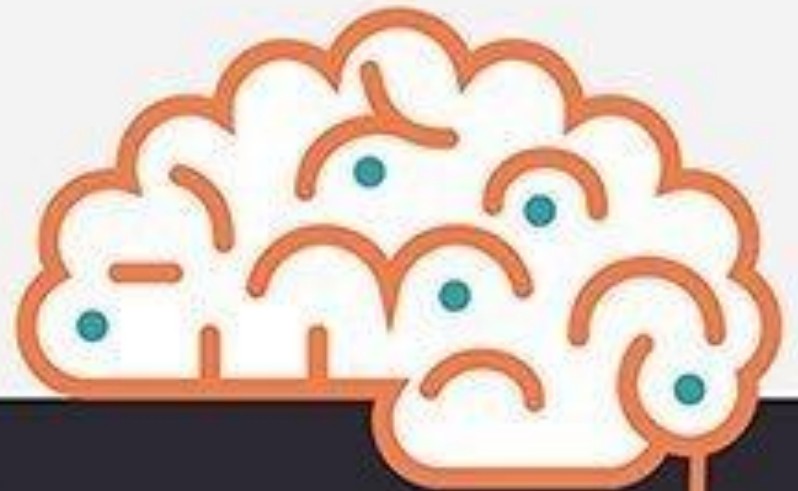
# Goals of AI

- ❑ Replicate Human Intelligence
- ❑ Solve Knowledge-intensive Tasks
- ❑ An Intelligent Connection of Perception and Action
- ❑ Building a Machine which can Perform Tasks that Requires Human Intelligence Such As:
  - ❑ Proving a Theorem
  - ❑ Playing Chess
  - ❑ Plan Some Surgical Operation
  - ❑ Driving a Car in Traffic

# What Contributes to AI?



# Components of AI



## Applications

- Image recognition
- Speech recognition
- Chatbots
- Natural language generation
- Sentiment analysis

## Types of models

- Deep learning
- Machine learning
- Neural networks

## Software/hardware for training and running models

- GPUs
- Parallel processing tools (like Spark)
- Cloud data storage and compute platforms

## Programming languages for building models

- Python
- TensorFlow
- Java
- C



# The Original 7 aspects of A.I. (1955)

1. Simulating higher functions of the human brain.
2. Programming computer to use general language.
3. Arranging hypothetical neurons in a manner so that they can form concepts.
4. A way to determine and measure problem complexity.
5. Self-improvement
6. Abstractions: Defined as the quality of dealing with ideas rather than events.
7. Randomness and creativity.

# Programming Without & With AI

| Programming Without AI                                                                       | Programming With AI                                                                                                                                                                                        |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A computer program without AI can answer the <b>specific</b> questions it is meant to solve. | A computer program with AI can answer the <b>generic</b> questions it is meant to solve.                                                                                                                   |
| Modification in the program leads to change in its structure.                                | AI programs can absorb new modifications by putting highly independent pieces of information together. Hence you can modify even a minute piece of information of program without affecting its structure. |
| Modification is not quick and easy. It may lead to affecting the program adversely.          | Quick and Easy program modification.                                                                                                                                                                       |

# Pros & Cons

- ❑ High Accuracy with Less

Errors:

- ❑ High-speed

- ❑ High Reliability

- ❑ Useful for Risky Areas

- ❑ Digital Assistant

- ❑ Useful as a Public Utility

- ❑ High Cost

- ❑ Can't Think out of The Box

- ❑ No Feelings and Emotions

- ❑ Increase Dependency on Machines:

- ❑ No Original Creativity

# Sub Areas of AI

- ❑ Game Playing
- ❑ Speech Recognition
- ❑ Computer Vision
- ❑ Expert Systems
- ❑ Mathematical Theorem Proving
- ❑ Natural Language Understanding
- ❑ Scheduling and Planning
- ❑ Artificial Neural Networks
- ❑ Machine Learning

# Application of AI

- ❑ Business; Financial Strategies
- ❑ Engineering: Check Design, Offer Suggestions to Create New Product, Expert Systems For ll Engineering Problems
- ❑ Manufacturing: Assembly, Inspection And Maintenance
- ❑ Medicine: Monitoring, Diagnosing
- ❑ Education: In Teaching
- ❑ Fraud Detection
- ❑ Object Identification
- ❑ Information Retrieval
- ❑ Space Shuttle Scheduling

# Future Scope of AI

- ❑ Cyber Security - Ensure In Curbing Hackers
- ❑ Data Analysis - Prediction
- ❑ Transport - Tesla
- ❑ Various Jobs - Robotic Process Automation
- ❑ Emotion Bots - Cortana & Alexa
- ❑ Marketing & Advertising – Flipkart

# 1.1 INTRODUCTION

| <b>INTELLIGENCE</b>                                                   | <b>ARTIFICIAL INTELLIGENCE</b>                                              |
|-----------------------------------------------------------------------|-----------------------------------------------------------------------------|
| It is a natural process.                                              | It is programmed by humans.                                                 |
| It is actually hereditary.                                            | It is not hereditary.                                                       |
| Knowledge is required for intelligence.                               | KB and electricity are required to generate output.                         |
| No human is an expert. We may get better solutions from other humans. | Expert systems are made which aggregate many person's experience and ideas. |

# 1.2 DEFINITION

- The study of how to make computers do things at which at the moment, people are better.
- “Artificial Intelligence is the ability of a computer to act like a human being”.
  - Systems that think like humans
  - Systems that act like humans
  - Systems that think rationally.
  - Systems that act rationally.

# Definitions of AI

## Thinking Humanly

“The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense.” (Haugeland, 1985)

“[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)

## Thinking Rationally

“The study of mental faculties through the use of computational models.”  
(Charniak and McDermott, 1985)

“The study of the computations that make it possible to perceive, reason, and act.”  
(Winston, 1992)

## Acting Humanly

“The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990)

“The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)

## Acting Rationally

“Computational Intelligence is the study of the design of intelligent agents.” (Poole *et al.*, 1998)

“AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

# Four Approaches of Artificial Intelligence:

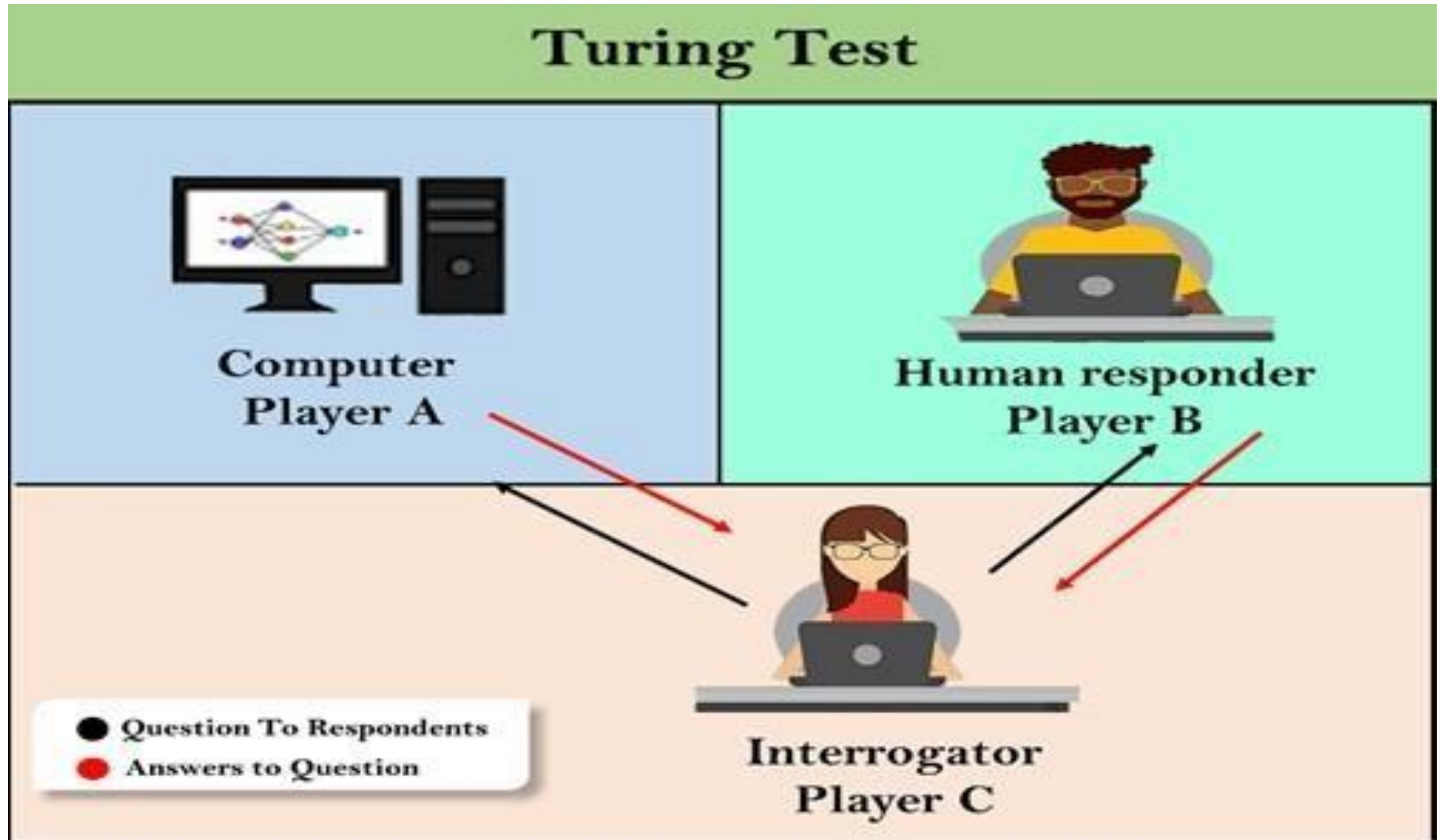
|        | Human- Like                                                                   | Rationally                                                                  |
|--------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Think: | <b>Cognitive Science Approach</b><br><i>“Machines that think like humans”</i> | <b>Laws of thought Approach</b><br><i>“ Machines that think Rationally”</i> |
| Act:   | <b>Turing Test Approach</b><br><i>“Machines that behave like humans”</i>      | <b>Rational Agent Approach</b><br><i>“Machines that behave Rationally”</i>  |

- a. **Intelligence** - Ability to apply knowledge in order to perform better in an environment.
- b. **Artificial Intelligence** - Study and construction of agent programs that perform well in a given environment, for a given agent architecture.
- c. **Agent** - An entity that takes action in response to precepts from an environment.
- d. **Rationality** - property of a system which does the “right thing” given what it knows.
- e. **Logical Reasoning** - A process of deriving new sentences from old, such that the new sentences are necessarily true if the old ones are true.

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH

- The Turing Test, proposed by Alan Turing (1950), was designed to provide a satisfactory operational definition of intelligence.
- A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH



**Turing Test**

CSA2001 AIML

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH

- natural language processing to enable it to communicate successfully in English;
- knowledge representation to store what it knows or hears;
- automated reasoning to use the stored information to answer questions and to draw new conclusions
- machine learning to adapt to new circumstances and to detect and extrapolate patterns.

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH

- **Total Turing Test** includes a video signal so that the interrogator can test the subject's perceptual abilities, as well as the opportunity for the interrogator to pass physical objects “through the hatch.”
- To pass the total Turing Test, the computer will need **computer vision** to perceive objects, and **robotics** to manipulate objects and move about.

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH

## Thinking humanly: The cognitive modelling approach

- Analyse how a given program thinks like a human, we must have some way of determining how humans think. The interdisciplinary field of cognitive science brings together computer models from AI and experimental techniques from psychology to try to construct precise and testable theories of the workings of the human mind.
- Although cognitive science is a fascinating field in itself, we are not going to be discussing it all that much in this book. We will occasionally comment on similarities or differences between AI techniques and human cognition. Real cognitive science, however, is necessarily based on experimental investigation of actual humans or animals, and we assume that the reader only has access to a computer for experimentation. We will simply note that AI and cognitive science continue to fertilize each other, especially in the areas of vision, natural language, and learning.

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH

## Thinking rationally: The “laws of thought” approach

- The Greek philosopher Aristotle was one of the first to attempt to codify “right thinking,” that is, irrefutable reasoning processes. His famous syllogisms provided patterns for argument structures that always gave correct conclusions given correct premises.
- For example, “Socrates is a man; all men are mortal; therefore Socrates is mortal.”
- These laws of thought were supposed to govern the operation of the mind, and initiated the field of **logic**.

# 1.3 ACTING HUMANLY: THE TURING TEST APPROACH

## Acting rationally: The rational agent approach

- Acting rationally means acting so as to achieve one's goals, given one's beliefs. An agent is just something that perceives and acts.
- The right thing: that which is expected to maximize goal achievement, given the available information
- Does not necessary involve thinking.
- For Example - blinking reflex- but should be in the service of rational action.

# 1.4 FUTURE OF ARTIFICIAL INTELLIGENCE

**Transportation:** Although it could take a decade or more to perfect them, autonomous cars will one day ferry us from place to place.

**Manufacturing:** AI powered robots work alongside humans to perform a limited range of tasks like assembly and stacking, and predictive analysis sensors keep equipment running smoothly.

**Healthcare:** In the comparatively AI-nascent field of healthcare, diseases are more quickly and accurately diagnosed, drug discovery is sped up and streamlined, virtual nursing assistants monitor patients and big data analysis helps to create a more personalized patient experience.

# 1.4 FUTURE OF ARTIFICIAL INTELLIGENCE

**Education:** Textbooks are digitized with the help of AI, early-stage virtual tutors assist human instructors and facial analysis gauges the emotions of students to help determine who's struggling or bored and better tailor the experience to their individual needs.

**Media:** Journalism is harnessing AI, too, and will continue to benefit from it. Bloomberg uses Cyborg technology to help make quick sense of complex financial reports. The Associated Press employs the natural language abilities of Automated Insights to produce 3,700 earning reports stories per year — nearly four times more than in the recent past

# 1.4 FUTURE OF ARTIFICIAL INTELLIGENCE

**Customer Service:** Last but hardly least, Google is working on an AI assistant that can place human-like calls to make appointments at, say, your neighborhood hair salon. In addition to words, the system understands context and tone.

# 1.5 CHARACTERISTICS OF INTELLIGENT AGENTS

- **Situatedness**

The agent receives some form of sensory input from its environment, and it performs some action that changes its environment in some way.

Examples of environments: the physical world and the Internet.

- **Autonomy**

The agent can act without direct intervention by humans or other agents and that it has control over its own actions and internal state.

# 1.5 CHARACTERISTICS OF INTELLIGENT AGENTS

- **Adaptivity**

1. The agent is capable of reacting flexibly to changes in its environment;
2. taking goal-directed initiative (i.e., is pro-active), when appropriate; and
3. Learning from its own experience, its environment, and interactions with others.

- **Sociability**

- The agent is capable of interacting in a peer-to-peer manner with other agents or humans

# 1.6 AGENTS AND ITS TYPES

- An agent is anything that can be viewed as perceiving its environment through sensors and acting upon that environment through actuators.

## **Human Sensors:**

- Eyes, ears, and other organs for sensors.

## **Human Actuators:**

- Hands, legs, mouth, and other body parts.

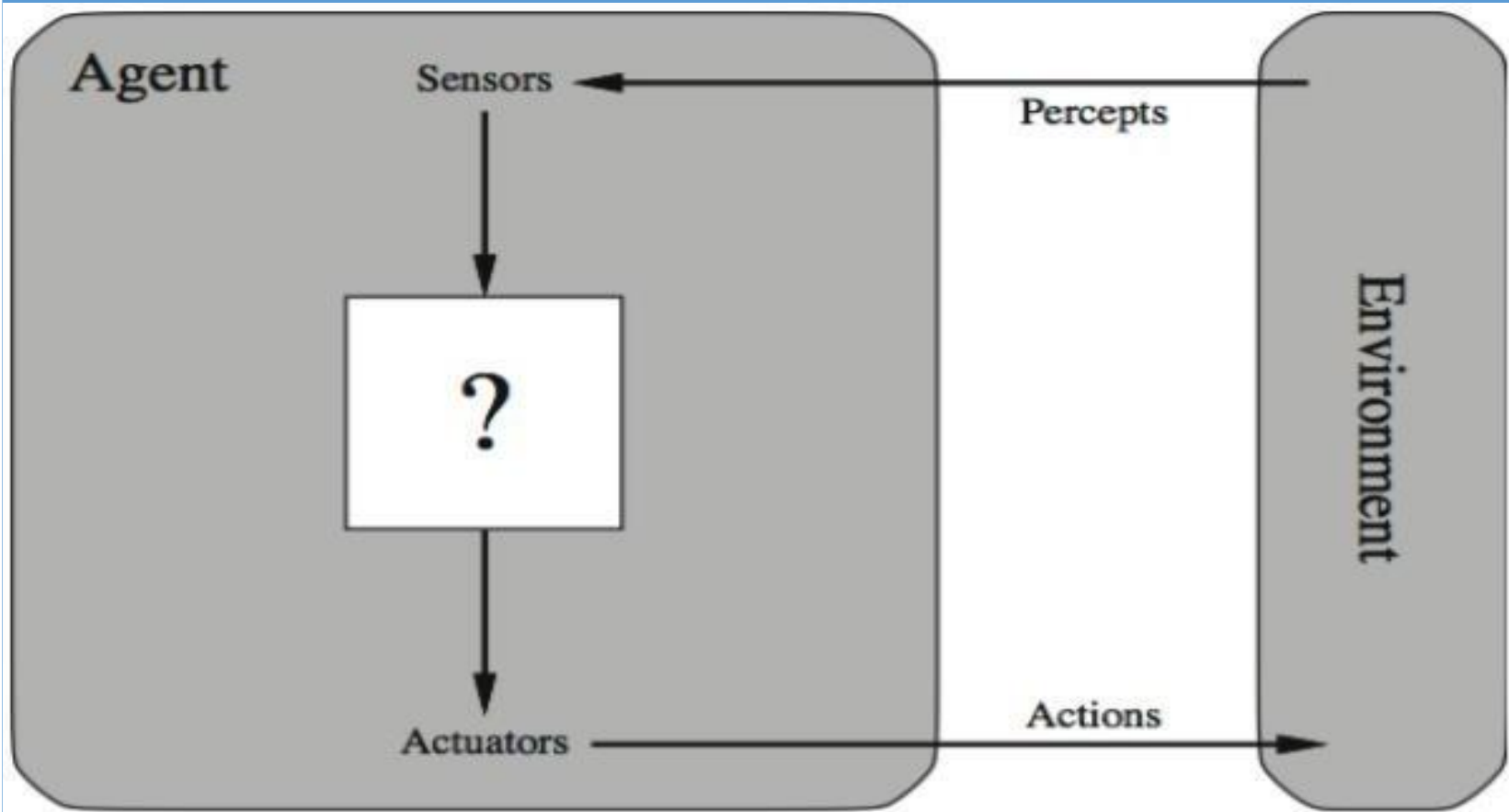
## **Robotic Sensors:**

- Mic, cameras and infrared range finders for sensors

## **Robotic Actuators:**

- Motors, Display, speakers etc

# 1.6 AGENTS AND ITS TYPES



**Agent types**

# 1.6 AGENTS AND ITS TYPES

An agent can be:

- **Human-Agent:** A human agent has eyes, ears, and other organs which work for sensors and hand, legs, vocal tract work for actuators.
- **Robotic Agent:** A robotic agent can have cameras, infrared range finder, NLP for sensors and various motors for actuators.
- **Software Agent:** Software agent can have keystrokes, file contents as sensory input and act on those inputs and display output on the screen.
- Hence the world around us is full of agents such as thermostat, cell phone, camera, and even we are also agents. Before moving forward, we should first know about sensors, effectors, and actuators.

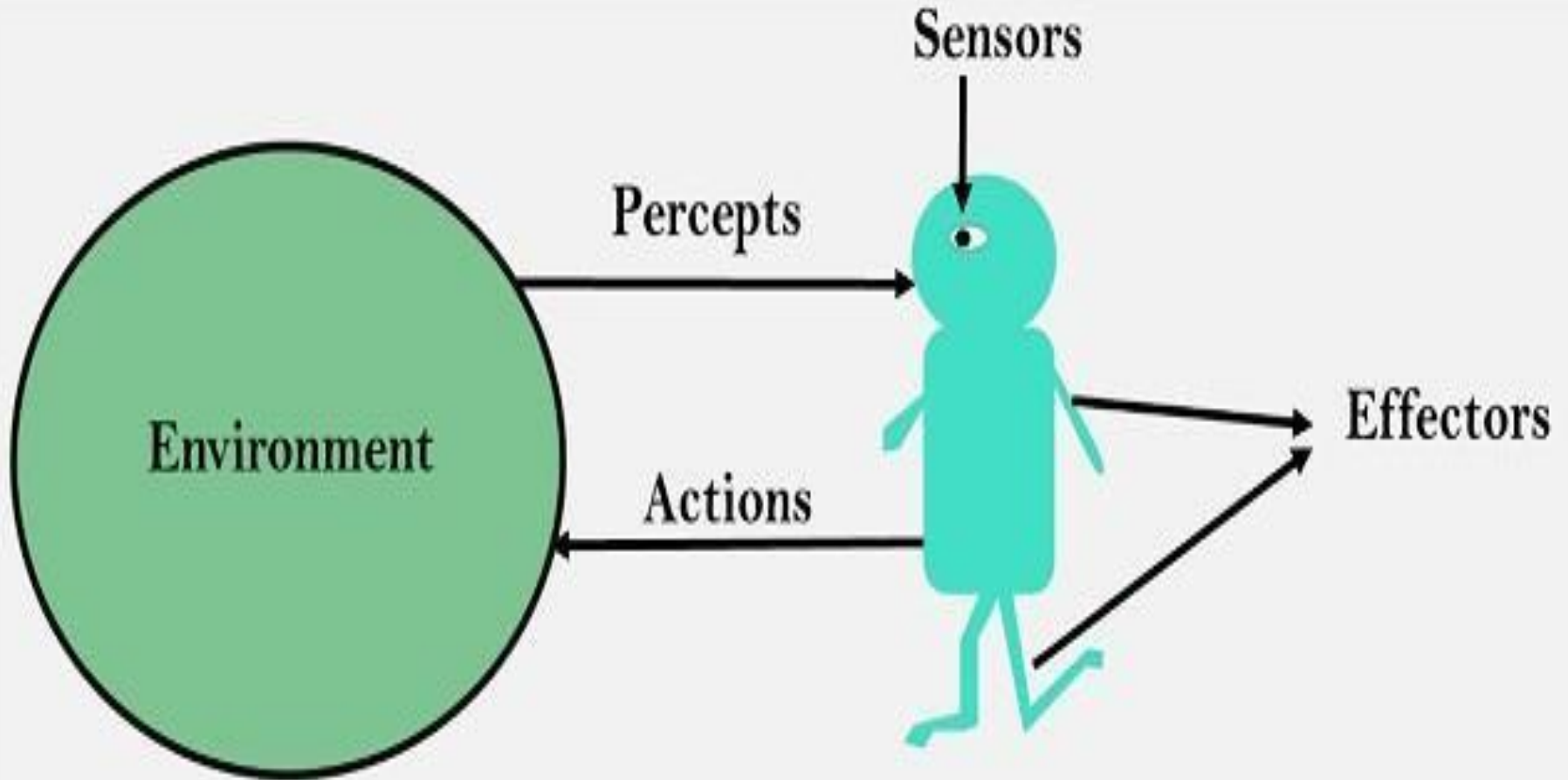
# 1.6 AGENTS AND ITS TYPES

**Sensor:** Sensor is a device which detects the change in the environment and sends the information to other electronic devices. An agent observes its environment through sensors.

**Actuators:** Actuators are the component of machines that converts energy into motion. The actuators are only responsible for moving and controlling a system. An actuator can be an electric motor, gears, rails, etc.

**Effectors:** Effectors are the devices which affect the environment. Effectors can be legs, wheels, arms, fingers, wings, fins, and display screen.

## 1.6 AGENTS AND ITS TYPES



**Effectors**

# 1.7 PROPERTIES OF ENVIRONMENT

- An **environment** is everything in the world which surrounds the agent, but it is not a part of an agent itself. An environment can be described as a situation in which an agent is present.
- The environment is where agent lives, operate and provide the agent with something to sense and act upon it.

# 1.7 PROPERTIES OF ENVIRONMENT

1. Fully observable vs Partially Observable
2. Deterministic vs Stochastic
3. Episodic vs Sequential
4. Single-agent vs Multi-agent
5. Static vs Dynamic
6. Discrete vs Continuous
7. Known vs Unknown
8. Accessible vs. Inaccessible

# 1.7 PROPERTIES OF ENVIRONMENT

## 1. Fully observable vs Partially Observable:

- If an agent sensor can sense or access the complete state of an environment at each point of time then it is a **fully observable** environment, else it is **partially observable**.
- A fully observable environment is easy as there is no need to maintain the internal state to keep track history of the world.
- An agent with no sensors in all environments then such an environment is called as unobservable.
- **Example:** chess – the board is fully observable, as are opponent's moves. Driving
  - – what is around the next bend is not observable and hence partially observable.

# 1.7 PROPERTIES OF ENVIRONMENT

## 2. Deterministic vs Stochastic

- If an agent's current state and selected action can completely determine the next state of the environment, then such environment is called a deterministic environment.
- A stochastic environment is random in nature and cannot be determined completely by an agent.
- In a deterministic, fully observable environment, agent does not need to worry about uncertainty.

## 3. Episodic vs Sequential

- In an episodic environment, there is a series of one-shot actions, and only the current percept is required for the action.
- However, in Sequential environment, an agent requires memory of past actions to determine the next best actions.

# 1.7 PROPERTIES OF ENVIRONMENT

## 4. Single-agent vs Multi-agent

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- However, if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
- The agent design problems in the multi-agent environment are different from single agent environment.

# 1.7 PROPERTIES OF ENVIRONMENT

## 5. Static vs Dynamic

- If the environment can change itself while an agent is deliberating then such environment is called a dynamic environment else it is called a static environment.
- Static environments are easy to deal because an agent does not need to continue looking at the world while deciding for an action.
- However for dynamic environment, agents need to keep looking at the world at each action.
- Taxi driving is an example of a dynamic environment whereas Crossword puzzles are an example of a static environment.

# 1.7 PROPERTIES OF ENVIRONMENT

## 6. Discrete vs Continuous

- If in an environment there are a finite number of precepts and actions that can be performed within it, then such an environment is called a discrete environment else it is called continuous environment.
- A chess game comes under discrete environment as there is a finite number of moves that can be performed.
- A self-driving car is an example of a continuous environment.

# 1.7 PROPERTIES OF ENVIRONMENT

## 7. Known vs Unknown

- Known and unknown are not actually a feature of an environment, but it is an agent's state of knowledge to perform an action.
- In a known environment, the results for all actions are known to the agent. While in unknown environment, agent needs to learn how it works in order to perform an action.
- It is quite possible that a known environment to be partially observable and an Unknown environment to be fully observable.

# 1.7 PROPERTIES OF ENVIRONMENT

## 8. Accessible vs. Inaccessible

- If an agent can obtain complete and accurate information about the state's environment, then such an environment is called an Accessible environment else it is called inaccessible.
- An empty room whose state can be defined by its temperature is an example of an accessible environment.
- Information about an event on earth is an example of Inaccessible environment.

# 1.7 PROPERTIES OF ENVIRONMENT

Task environments, which are essentially the "problems" to which rational agents are the "solutions."

**PEAS:** Performance Measure, Environment, Actuators, Sensors

## Performance

- The output which we get from the agent. All the necessary results that an agent gives after processing comes under its performance.

## Environment

- All the surrounding things and conditions of an agent fall in this section. It basically consists of all the things under which the agents work.

# 1.7 PROPERTIES OF ENVIRONMENT

## Actuators

- The devices, hardware or software through which the agent performs any actions or processes any information to produce a result are the actuators of the agent.

## Sensors

- The devices through which the agent observes and perceives its environment are the sensors of the agent.

# 1.7 PROPERTIES OF ENVIRONMENT

| Agent Type                      | Performance Measure                 | Environment                      | Actuators                                                     | Sensors                                                 |
|---------------------------------|-------------------------------------|----------------------------------|---------------------------------------------------------------|---------------------------------------------------------|
| Medical diagnosis system        | Healthy patient, reduced costs      | Patient, hospital, staff         | Display of questions, tests, diagnoses, treatments, referrals | Keyboard entry of symptoms, findings, patient's answers |
| Satellite image analysis system | Correct image categorization        | Downlink from orbiting satellite | Display of scene categorization                               | Color pixel arrays                                      |
| Part-picking robot              | Percentage of parts in correct bins | Conveyor belt with parts; bins   | Jointed arm and hand                                          | Camera, joint angle sensors                             |
| Refinery controller             | Purity, yield, safety               | Refinery, operators              | Valves, pumps, heaters, displays                              | Temperature, pressure, chemical sensors                 |
| Interactive English tutor       | Student's score on test             | Set of students, testing agency  | Display of exercises, suggestions, corrections                | Keyboard entry                                          |

**Examples of agent types and their PEAS descriptions**

# 1.7 PROPERTIES OF ENVIRONMENT

**Rational Agent** - A system is rational if it does the “right thing”.  
Given what it knows.

## **Characteristic of Rational Agent**

- The agent's prior knowledge of the environment.
- The performance measure that defines the criterion of success.
- The actions that the agent can perform.
- The agent's percept sequence to date.

For every possible percept sequence, a rational agent should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

# 1.7 PROPERTIES OF ENVIRONMENT

- An **omniscient agent** knows the actual outcome of its actions and can act accordingly; but omniscience is impossible in reality.
- Ideal **Rational Agent** precepts and does things. It has a greater performance measure.
  - Eg. Crossing road. Here first perception occurs on both sides and then only action. No perception occurs in **Degenerate Agent**.
  - Eg. Clock. It does not view the surroundings. No matter what happens outside. The clock works based on inbuilt program.

# 1.7 PROPERTIES OF ENVIRONMENT

- **Ideal Agent** describes by ideal mappings. “Specifying which action an agent ought to take in response to any given percept sequence provides a design for ideal agent”.
  - Eg. SQRT function calculation in calculator.
- Doing actions in order to modify future precepts-sometimes called information gathering- is an important part of rationality.
- A rational agent should be autonomous-it should learn from its own prior knowledge (experience).

# 1.7 PROPERTIES OF ENVIRONMENT

## The Structure of Intelligent Agents

Agent = Architecture + Agent Program

Architecture = the machinery that an agent executes on.  
(Hardware)

Agent Program = an implementation of an  
agent function. (Algorithm, Logic – Software)

# 1.8 TYPES OF AGENTS

Agents can be grouped into five classes based on their degree of perceived intelligence and capability :

1. Simple Reflex Agents
2. Model-Based Reflex Agents
3. Goal-Based Agents
4. Utility-Based Agents
5. Learning Agent

# 1.8 TYPES OF AGENTS

## 1. The Simple reflex agents

- The Simple reflex agents are the simplest agents. These agents take decisions on the basis of the current percepts and ignore the rest of the percept history (past State).
- These agents only succeed in the fully observable environment.
- The Simple reflex agent does not consider any part of percepts history during their decision and action process.
- The Simple reflex agent works on Condition-action rule, which means it maps the current state to action. Such as a Room Cleaner agent, it works only if there is dirt in the room.

# 1.8 TYPES OF AGENTS

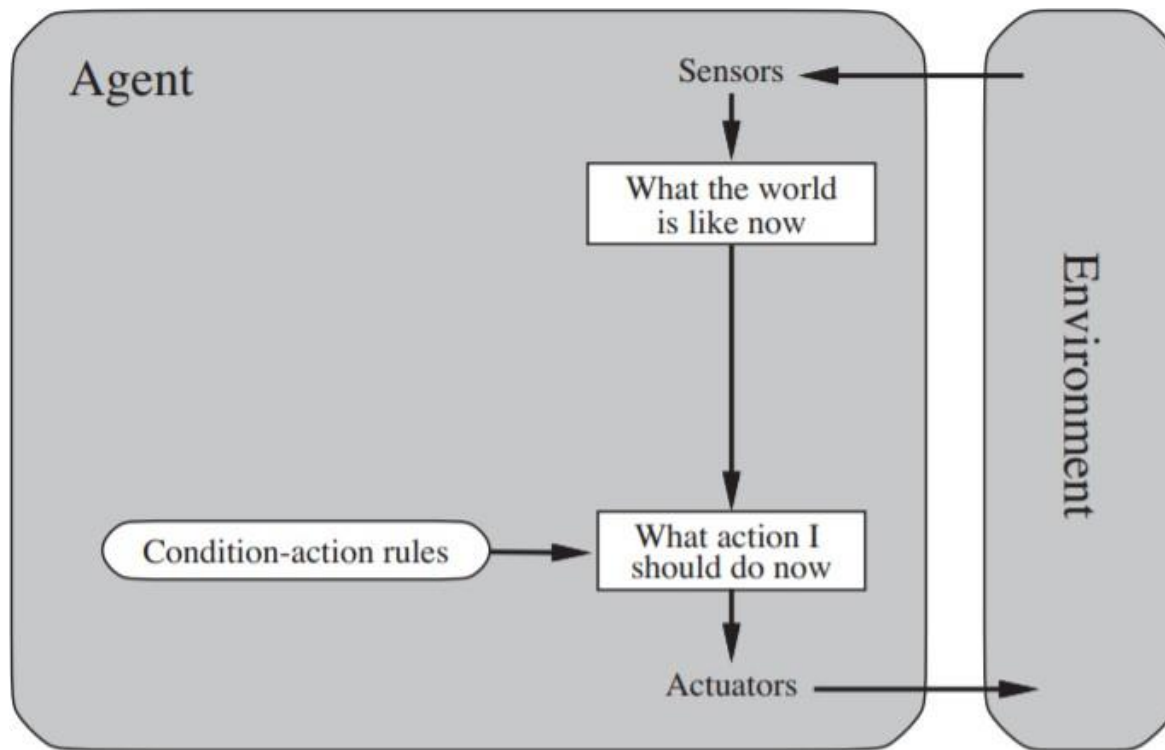
## 1. The Simple reflex agents

- Problems for the simple reflex agent design approach:
  - ☐ They have very limited intelligence
  - ☐ They do not have knowledge of non-perceptual parts of the current state
  - ☐ Mostly too big to generate and to store.
  - ☐ Not adaptive to changes in the environment.

# 1.8 TYPES OF AGENTS

## 1. The Simple reflex agents

- **Condition-Action Rule** – It is a rule that maps a state (condition) to an action.
  - Ex: if car-in-front-is-braking then initiate- braking.



**A simple reflex agent**

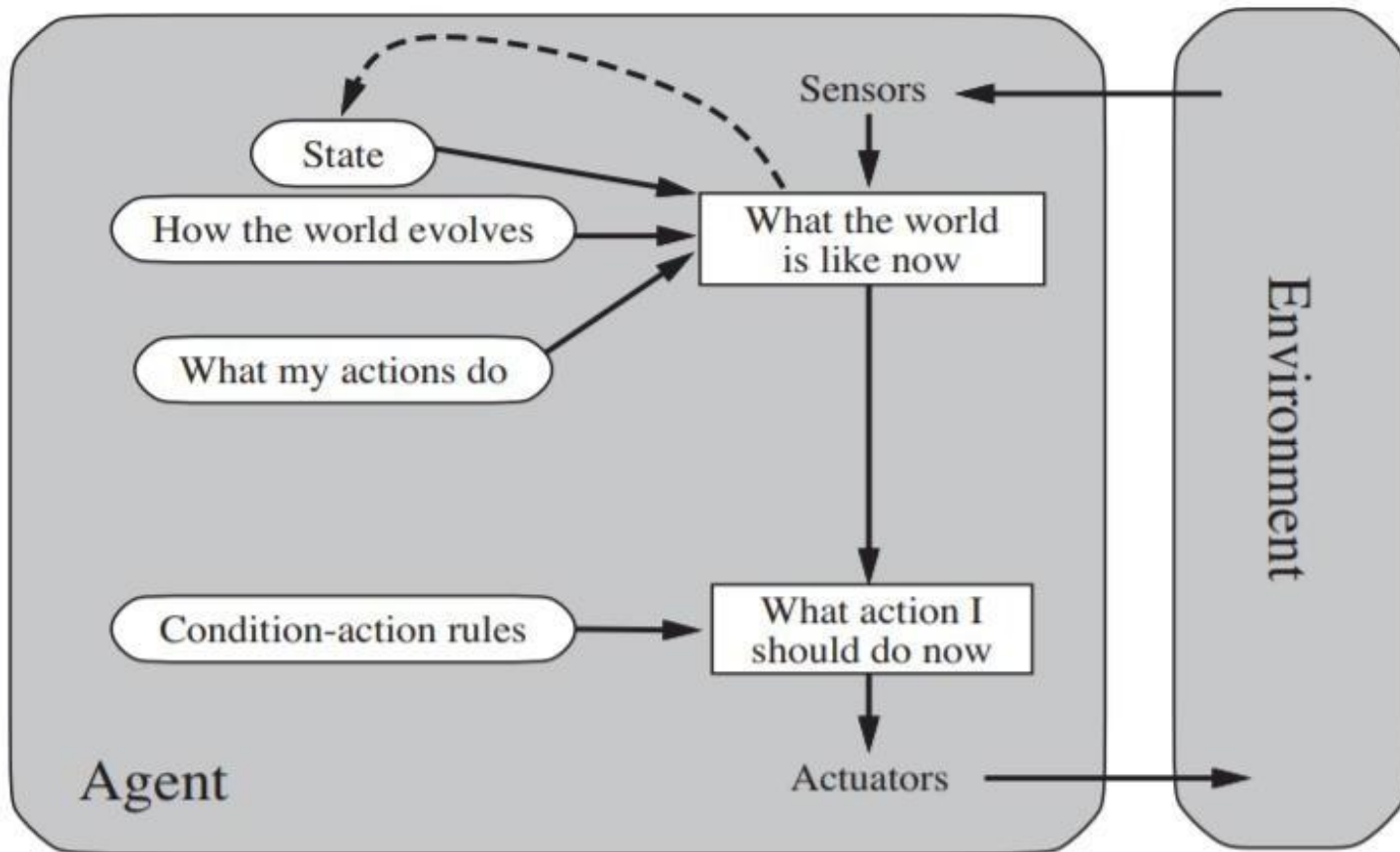
# 1.8 TYPES OF AGENTS

## 2. Model Based Reflex Agents

- The Model-based agent can work in a partially observable environment, and track the situation.
- A model-based agent has two important factors:
  - **Model:** It is knowledge about "how things happen in the world," so it is called a Model-based agent.
  - **Internal State:** It is a representation of the current state based on percept history.
- These agents have the model, "which is knowledge of the world" and based on the model they perform actions.
- Updating the agent state requires information about:
  - How the world evolves
  - How the agent's action affects the world.

# 1.8 TYPES OF AGENTS

## 2. Model Based Reflex Agents



**A model-based reflex agent**

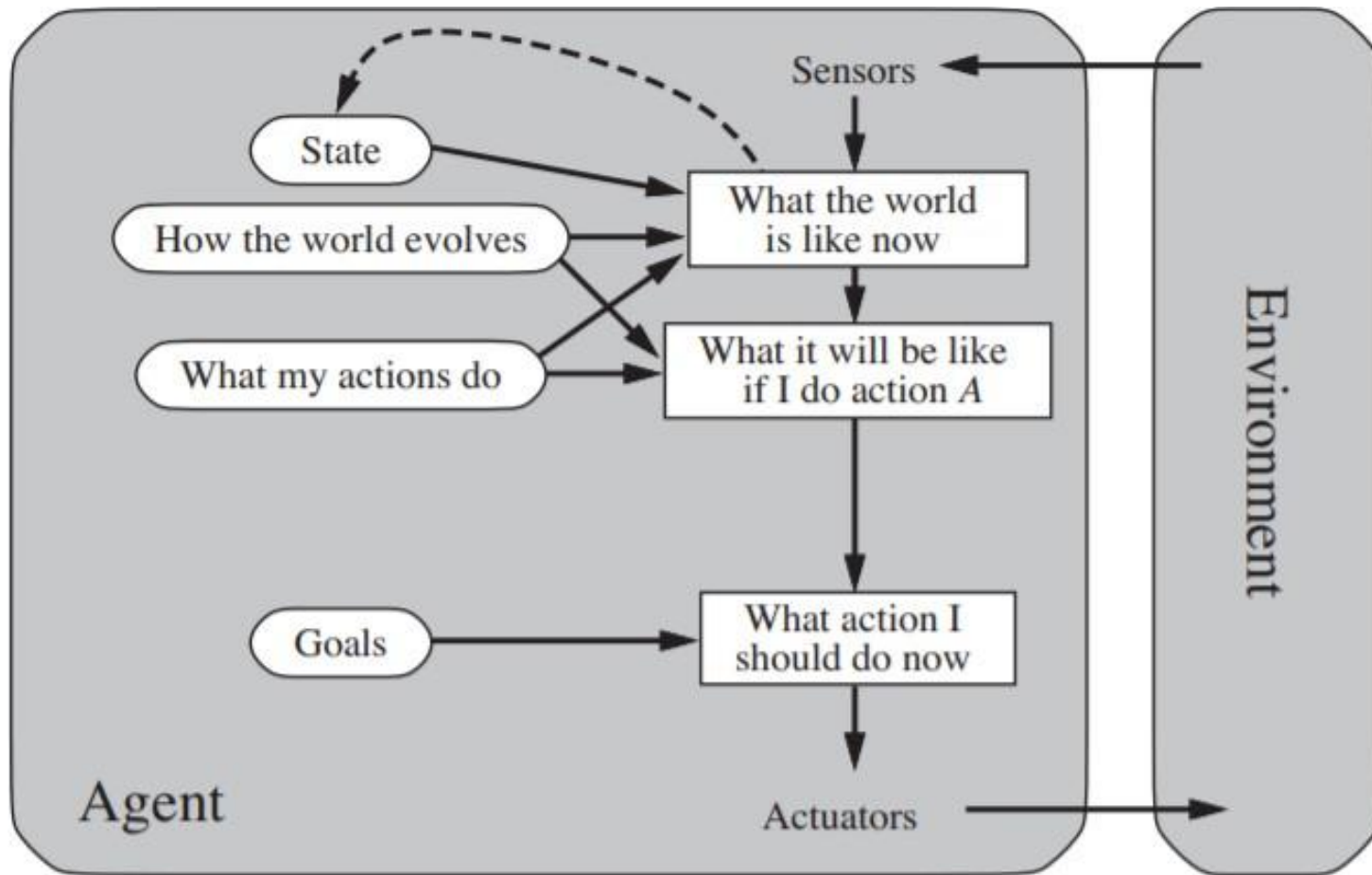
# 1.8 TYPES OF AGENTS

## 3. Goal Based Agents

- The knowledge of the current state environment is not always sufficient to decide for an agent to what to do.
- The agent needs to know its goal which describes desirable situations.
- Goal-based agents expand the capabilities of the model-based agent by having the "goal" information.
- They choose an action, so that they can achieve the goal.
- These agents may have to consider a long sequence of possible actions before deciding whether the goal is achieved or not. Such considerations of different scenario are called searching and planning, which makes an agent proactive.

# 1.8 TYPES OF AGENTS

## 3. Goal Based Agents



**A goal-based agent**

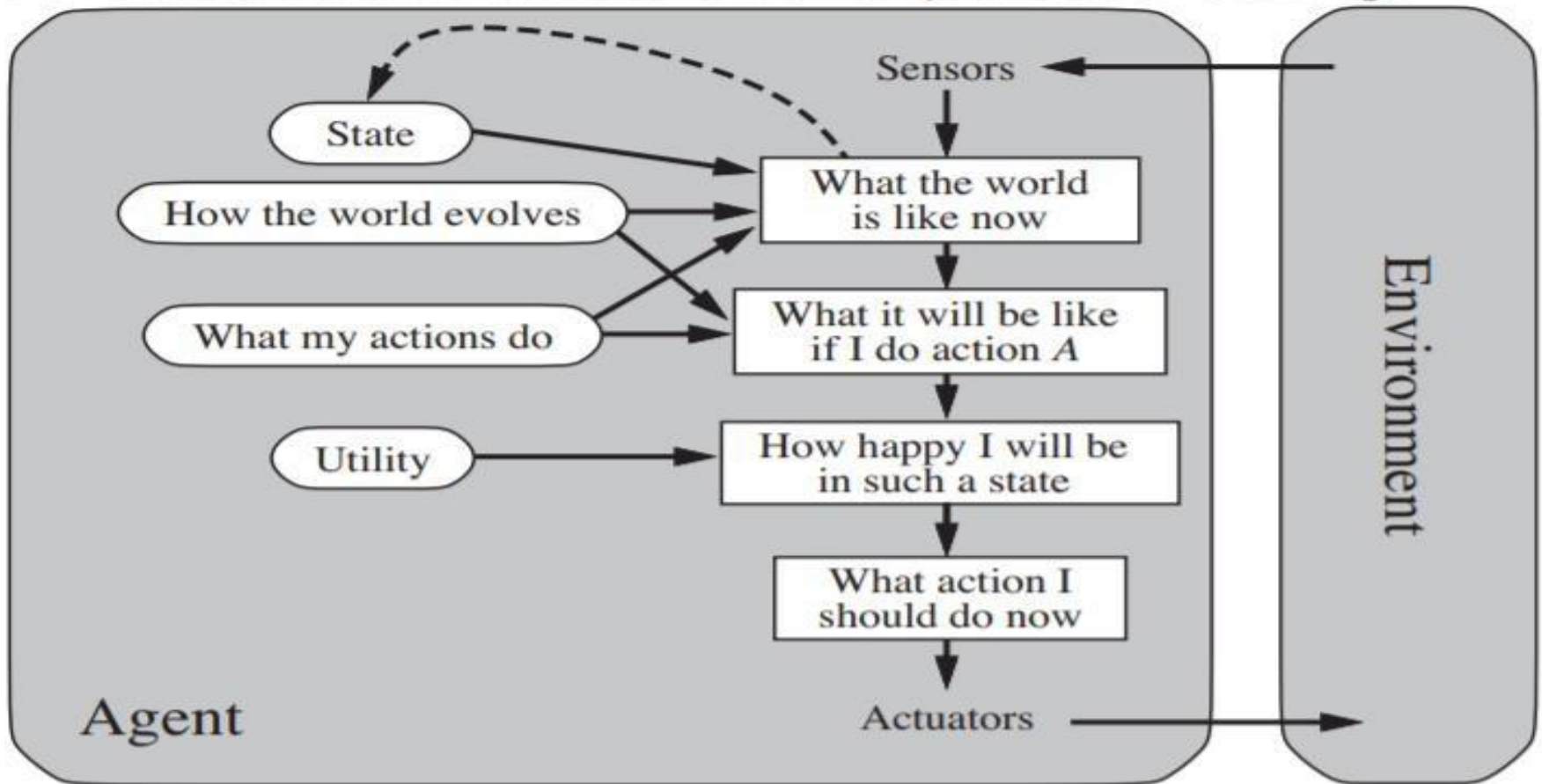
# 1.8 TYPES OF AGENTS

## 4. Utility Based Agents

- These agents are similar to the goal-based agent but provide an extra component of utility measurement (“Level of Happiness”) which makes them different by providing a measure of success at a given state.
- Utility-based agent act based not only goals but also the best way to achieve the goal.
- The Utility-based agent is useful when there are multiple possible alternatives, and an agent has to choose in order to perform the best action.
- The utility function maps each state to a real number to check how efficiently each action achieves the goals.

# 1.8 TYPES OF AGENTS

## 4. Utility Based Agents



**A utility-based agent**

# 1.8 TYPES OF AGENTS

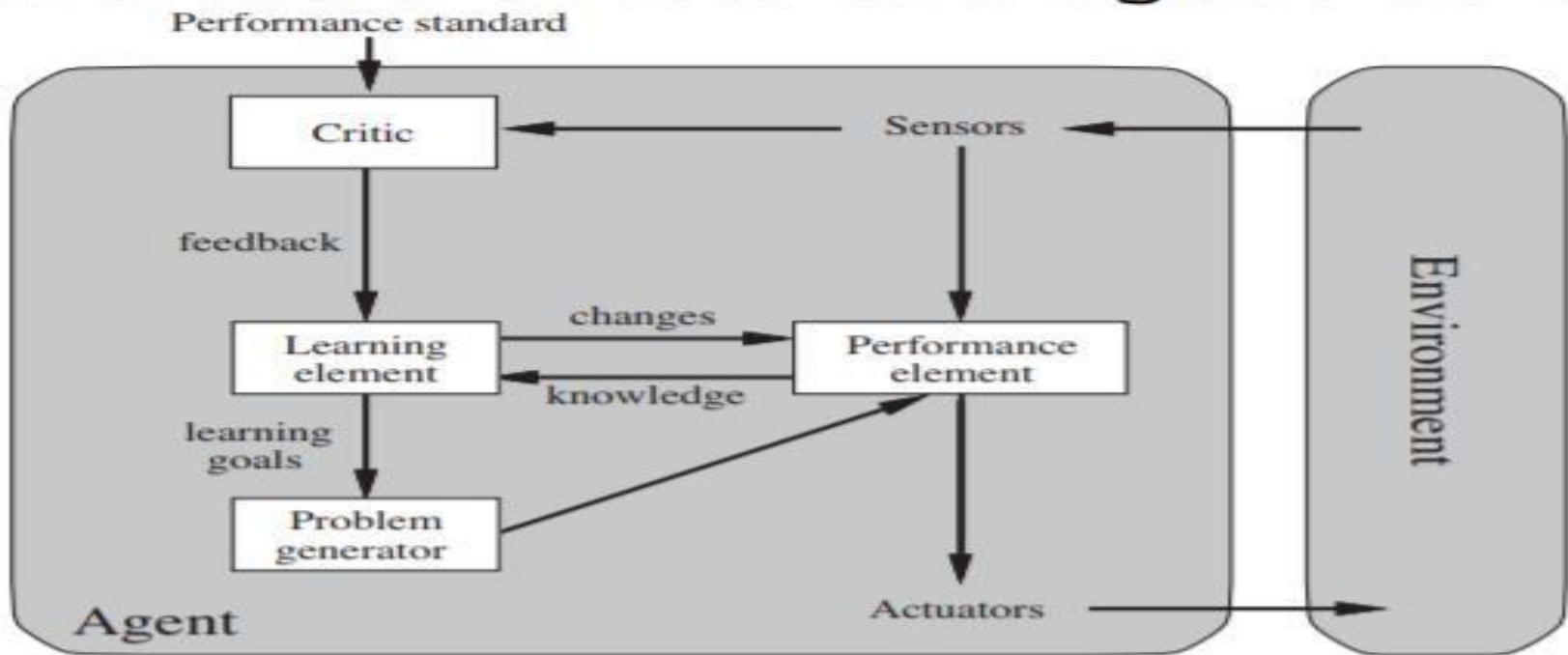
## 5. Learning Agents

- A learning agent in AI is the type of agent which can learn from its past experiences, or it has learning capabilities.
- It starts to act with basic knowledge and then able to act and adapt automatically through learning.
- A learning agent has mainly four conceptual components, which are:
  - a. **Learning element:** It is responsible for making improvements by learning from environment
  - b. **Critic:** Learning element takes feedback from critic which describes that how well the agent is doing with respect to a fixed performance standard.
  - c. **Performance element:** It is responsible for selecting external action
  - d. **Problem generator:** This component is responsible for suggesting actions that will lead to new and informative experiences.

# 1.8 TYPES OF AGENTS

## 5. Learning Agents

- Hence, learning agents are able to learn, analyze performance, and look for new ways to improve the performance.



## Learning Agents

# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

## Problem-solving agents

In Artificial Intelligence, Search techniques are universal problem-solving methods. **Rational agents or Problem-solving agents** in AI mostly used these search strategies or algorithms to solve a specific problem and provide the best result. Problem-solving agents are the goal-based agents and use atomic representation. In this topic, we will learn various problem-solving search algorithms.

- Some of the most popularly used problem solving with the help of artificial intelligence are:

1. Chess.
2. Travelling Salesman Problem.
3. Tower of Hanoi Problem.
4. Water-Jug Problem.
5. N-Queen Problem.

# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

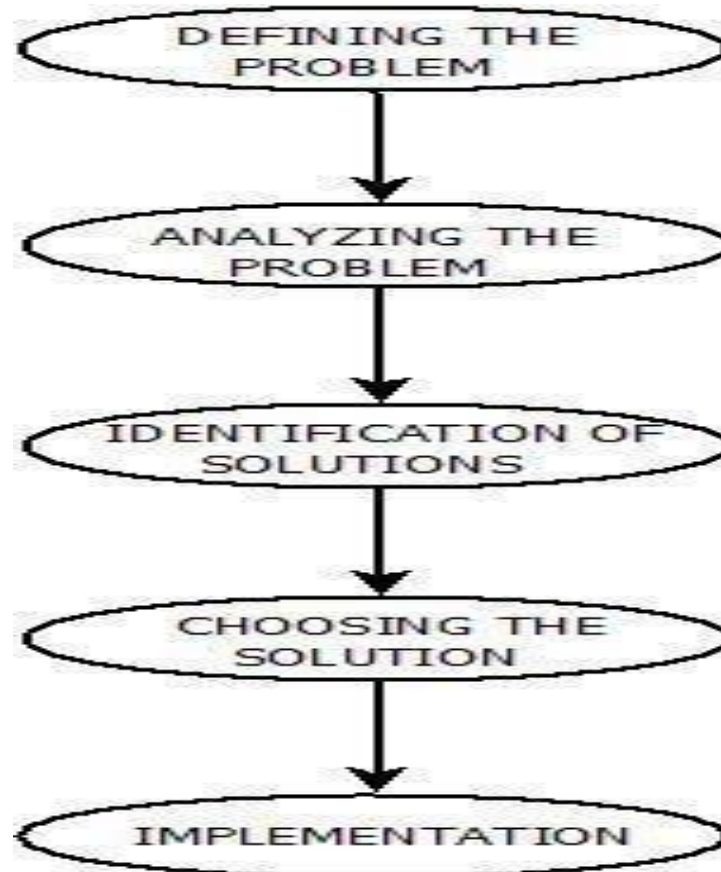
## Problem Searching

- In general, searching refers to as finding information one needs.
- Searching is the most commonly used technique of problem solving in artificial intelligence.
- The searching algorithm helps us to search for solution of particular problem.
- **Problem:** Problems are the issues which comes across any system. A solution is needed to solve that particular problem.

# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

## Steps : Solve Problem Using Artificial Intelligence

- The process of solving a problem consists of five steps. These are:



# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

**Defining The Problem:** The definition of the problem must be included precisely. It should contain the possible initial as well as final situations which should result in acceptable solution.

- 1. Analyzing The Problem:** Analyzing the problem and its requirement must be done as few features can have immense impact on the resulting solution.
- 2. Identification of Solutions:** This phase generates reasonable amount of solutions to the given problem in a particular range.
- 3. Choosing a Solution:** From all the identified solutions, the best solution is chosen basis on the results produced by respective solutions.
- 4. Implementation:** After choosing the best solution, its implementation is done.

# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

## Measuring problem-solving performance

We can evaluate an algorithm's performance in four ways:

**Completeness:** Is the algorithm guaranteed to find a solution when there is one?

**Optimality:** Does the strategy find the optimal solution?

**Time complexity:** How long does it take to find a solution?

**Space complexity:** How much memory is needed to perform the search?

# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

## Search Algorithm Terminologies

- **Search:** Searching is a step by step procedure to solve a search-problem in a given search space. A search problem can have three main factors:
  1. **Search Space:** Search space represents a set of possible solutions, which a system may have.
  2. **Start State:** It is a state from where agent begins the search.
  3. **Goal test:** It is a function which observe the current state and returns whether the goal state is achieved or not.
- **Search tree:** A tree representation of search problem is called Search tree. The root of the search tree is the root node which is corresponding to the initial state.
- **Actions:** It gives the description of all the available actions to the agent.

# 1.9 PROBLEM SOLVING APPROACH TO TYPICAL AI PROBLEMS

- **Transition model:** A description of what each action do, can be represented as a transition model.
- **Path Cost:** It is a function which assigns a numeric cost to each path.
- **Solution:** It is an action sequence which leads from the start node to the goal node. **Optimal Solution:** If a solution has the lowest cost among all solutions.

# Example Problems

- A **Toy Problem** is intended to illustrate or exercise various problem-solving methods.
- A **real- world problem** is one whose solutions people actually care about.

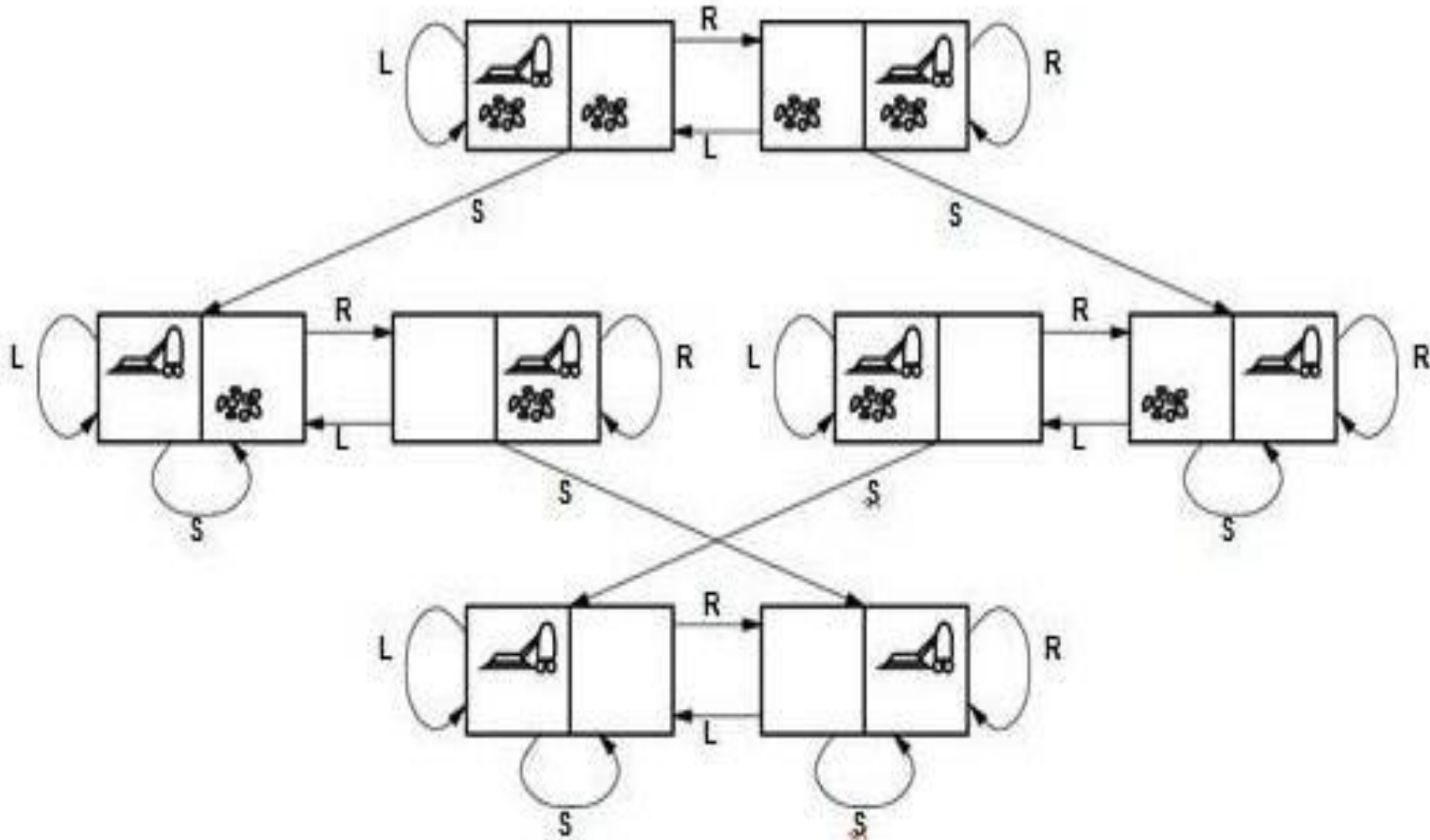
## Toy Problems

- Vacuum World
- **States:** The state is determined by both the agent location and the dirt locations. The agent is in one of the 2 locations, each of which might or might not contain dirt. Thus there are  $2 \times 2^2 = 8$  possible world states.
- **Initial state:** Any state can be designated as the initial state.
- **Actions:** In this simple environment, each state has just three actions: Left, Right, and Suck. Larger environments might also include Up and Down.

# Example Problems

- **Transition model:** The actions have their expected effects, except that moving Left in the leftmost square, moving Right in the rightmost square, and Sucking in a clean square have no effect.
- **Goal test:** This checks whether all the squares are clean.
- **Path cost:** Each step costs 1, so the path cost is the number of steps in the path.

# Example Problems



Vacuum World State Space Graph

# 1) 8- Puzzle Problem

|   |   |   |
|---|---|---|
| 7 | 2 | 4 |
| 5 |   | 6 |
| 8 | 3 | 1 |

Start State

|   |   |   |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 |   |

Goal State

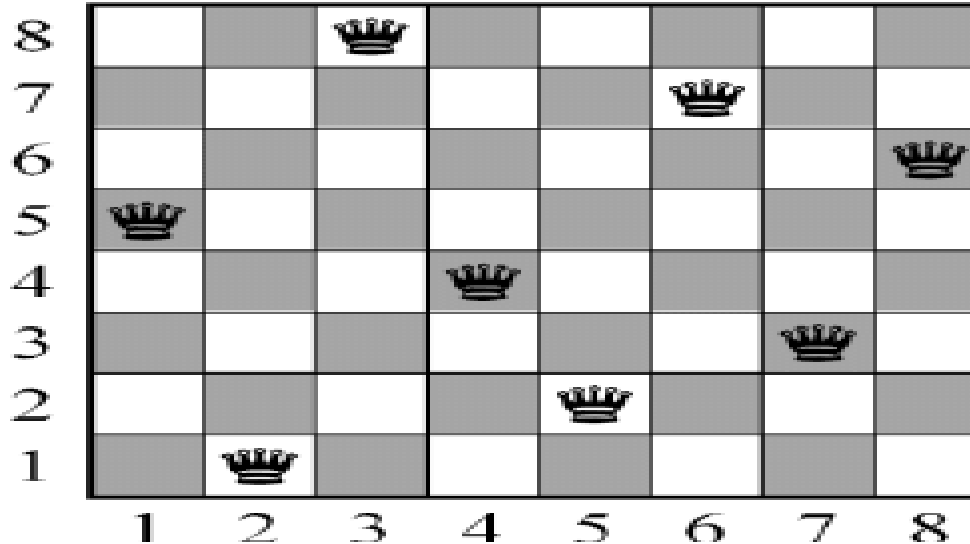
## 8- Puzzle Problem

- **States:** A state description specifies the location of each of the eight tiles and the blank in one of the nine squares.

# 1) 8- Puzzle Problem

- **Initial state:** Any state can be designated as the initial state. Note that any given goal can be reached from exactly half of the possible initial states.
- The simplest formulation defines the actions as movements of the blank space Left, Right, Up, or Down. Different subsets of these are possible depending on where the blank is.
- **Transition model:** Given a state and action, this returns the resulting state; for example, if we apply Left to the start state, the resulting state has the 5 and the blank switched.
- **Goal test:** This checks whether the state matches the goal configuration shown in Figure. Path cost: Each step costs 1, so the path cost is the number of steps in the path.

## 2) Queens Problem



### Queens Problem

- **States:** Any arrangement of 0 to 8 queens on the board is a state.
- **Initial state:** No queens on the board.
- **Actions:** Add a queen to any empty square.
- **Transition model:** Returns the board with a queen added to the specified square.
- **Goal test:** 8 queens are on the board, none attacked.

### 3) Water Jug Problem

- Consider the water jug problem: You are given two jugs, a 4-gallon one and 3-gallon one. Neither has any measuring marker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallon of water from the 4-gallon jug ?
- Explicit Assumptions: A jug can be filled from the pump, water can be poured out of a jug on to the ground, water can be poured from one jug to another and that there are no other measuring devices available.
- Here the initial state is  $(0, 0)$ . The goal state is  $(2, n)$  for any value of  $n$ .

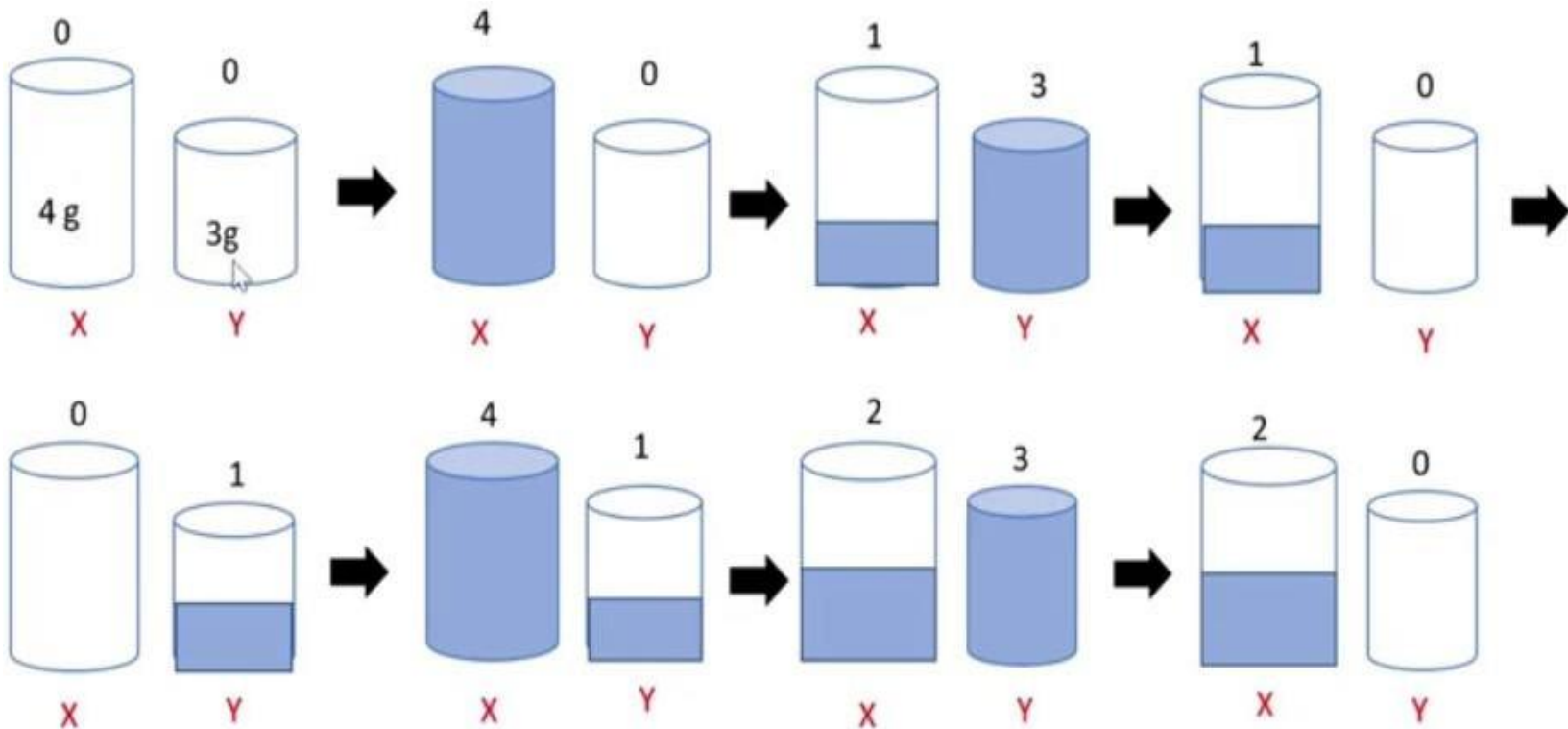
### 3) Water Jug Problem

- **State Space Representation:** we will represent a state of the problem as a tuple  $(x, y)$  where  $x$  represents the amount of water in the 4-gallon jug and  $y$  represents the amount of water in the 3-gallon jug. Note that  $0 \leq x \leq 4$ , and  $0 \leq y \leq 3$ .
- To solve this we have to make some assumptions not mentioned in the problem. They are:
  - We can fill a jug from the pump.
  - We can pour water out of a jug to the ground.
  - We can pour water from one jug to another.
  - There is no measuring device available.
- Operators - we must define a set of operators that will take us from one state to another.

### 3) Water Jug Problem

| Sr . | Current State                            | Next State         | Descriptions                                                                          |
|------|------------------------------------------|--------------------|---------------------------------------------------------------------------------------|
| 1    | $(x,y)$ if $x < 4$                       | $(4,y)$            | Fill the 4 gallon jug                                                                 |
| 2    | $(x,y)$ if $x < 3$                       | $(x,3)$            | Fill the 3 gallon jug                                                                 |
| 3    | $(x,y)$ if $x > 0$                       | $(x - d, y)$       | Pour some water out of the 4 gallon jug                                               |
| 4    | $(x,y)$ if $y > 0$                       | $(x, y - d)$       | Pour some water out of the 3 gallon jug                                               |
| 5    | $(x,y)$ if $y > 0$                       | $(0, y)$           | Empty the 4 gallon jug                                                                |
| 6    | $(x,y)$ if $y > 0$                       | $(x, 0)$           | Empty the 3 gallon jug on the ground                                                  |
| 7    | $(x,y)$ if $x + y \geq 4$<br>and $y > 0$ | $(4, y - (4 - x))$ | Pour water from the 3 gallon jug into the 4 gallon jug until the 4 gallon jug is full |
| 8    | $(x,y)$ if $x + y \geq 3$<br>and $x > 0$ | $(x - (3 - x), 3)$ | Pour water from the 4 gallon jug into the 3 gallon jug until the 3 gallon jug is full |
| 9    | $(x,y)$ if $x + y \leq 4$<br>and $y > 0$ | $(x + y, 0)$       | Pour all the water from the 3 gallon jug into the 4 gallon jug                        |
| 10   | $(x,y)$ if $x + y \leq 3$<br>and $x > 0$ | $(0, x + y)$       | Pour all the water from the 4 gallon jug into the 3 gallon jug                        |
| 11   | $(0, 2)$                                 | $(2, 0)$           | Pour the 2 gallons from 3 gallon jug into the 4 gallon jug                            |
| 12   | $(2, y)$                                 | $(0, y)$           | Empty the 2 gallons in the 4 gallon jug on the ground                                 |

### 3) Water Jug Problem



**Solution**

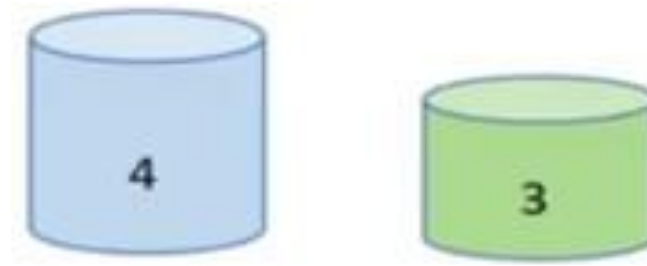
### 3) Water Jug Problem

#### Solution

| S.No . | Gallons in 4-gel jug(x) | Gallons in 3-gel jug (y) | Rule Applied             |
|--------|-------------------------|--------------------------|--------------------------|
| 1.     | 0                       | 0                        | Initial state            |
| 2..    | 4                       | 0                        | 1. Fill 4                |
| 3      | 1                       | 3                        | 6. Poor 4 into 3 to fill |
| 4.     | 1                       | 0                        | 4. Empty 3               |
| 5.     | 0                       | 1                        | 8. Poor all of 4 into 3  |
| 6.     | 4                       | 1                        | 1. Fill 4                |
| 7.     | 2                       | 3                        | 6. Poor 4 into 3         |

### 3) Water Jug Problem

- 4-gallon one and a 3-gallon Jug



- No measuring mark on the jug.
- There is a pump to fill the jugs with water.
- How can you get exactly 2 gallon of water into the 4-gallon jug?

**END**